AIN SHAMS UNIVERSITY
FACULTY OF ENGINEERING
MECHATRONICS ENGINEERING DEPARTMENT

Fall 2025

CSE480: Machine Vision

Major Task

Objective

In this project, students will design and implement a deep learning–based computer vision system that can recognize human actions and facial emotions from a live camera feed in real time.

The system will integrate Convolutional Neural Networks (CNNs) for spatial feature extraction and Long Short-Term Memory (LSTM) networks for temporal pattern recognition. This hybrid CNN–LSTM architecture will enable the model to analyze motion sequences and facial expressions over consecutive frames to classify ongoing human behavior and emotion. By completing both milestones, students will build an end-to-end pipeline capable of detecting, classifying, and visualizing human behavior through real-time video analysis.

Milestone 1: Dataset Preparation and Model Development

Deadline: Week 11

1. Data Collection and Preprocessing:
    - Select or prepare datasets for each task:
        - Action recognition: UCF-101 or custom video dataset with simple actions (walking, standing, waving, sitting).
        - Emotion recognition: FER-2013 dataset.
    - Extract video frames and preprocess them (resizing, normalization, and augmentation).
    - For the action recognition model, prepare short frame sequences (e.g., 10–20 consecutive frames per clip).
2. Model Design:
    - Action Recognition Model:
        - Use a CNN (e.g., ResNet or MobileNet) for spatial feature extraction.
        - Feed extracted features into an LSTM network, 1-2 layers, to capture temporal dependencies and recognize actions over time.
3. Training and Optimization:
    - Train each model using different optimization algorithms and compare their performance:
        - Stochastic Gradient Descent (SGD)
        - Adam
        - Adagrad
    - Record training and validation accuracy, training time, and loss evolution for each optimizer.

Milestone 2: System Integration and Real-Time Implementation

Deadline: Week 14

1. Model Design
   - Emotion Recognition Model:
     - Use a CNN-based classifier (e.g., VGG, ResNet, or EfficientNet) to recognize facial emotions from individual frames
2. Model Integration with Camera Feed:
   - Combine the trained CNN–LSTM action recognition model and the CNN-based emotion recognition model into a unified real-time pipeline.
   - Capture live frames from the webcam using OpenCV and process them for detection and classification.
   - Overlay the predicted action and emotion labels on the live video feed.
3. Performance Evaluation:
   - Evaluate accuracy on unseen test clips.
   - Measure frame processing rate (FPS) and latency for real-time predictions.
   - Compare the accuracy and responsiveness of different architectures and optimizers.

Tools and Libraries:

- Python (with TensorFlow/Keras, OpenCV, NumPy, Matplotlib, Pandas)
- Any suitable IDE (PyCharm, Jupyter Notebook, etc.)

Submission:

Students must submit a neat report for each milestone, including:

- Problem Definition and Importance (1 Page).
- Data Cleaning and Preprocessing Steps (1 Page).
- Methods and Algorithms (2-3 Pages).
- Experimental Results (sample outputs, model comparison, and analysis).
- Discussion on Model Performance with Justification.
- Appendix with Code.

Warnings:

- Plagiarism is prohibited.
- Assignments with no reports or no presentations will not be graded.